

PALANTIR FOUNDRY PROJECT ENABLEMENT PROGRAM

Day 1 Study Material

Project Architecture & Solution Walkthrough

Phase 1 — Foundation Accelerator

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1. Session Overview & Objectives

Day 1 opens the Foundation Accelerator phase of the engagement. Before any hands-on development begins, the whole team needs a shared, accurate picture of how Foundry is architected and how the pieces you will build over the next ten days fit together.

By the end of today's session, you will be able to:

- Describe the major architectural layers of the Foundry platform
- Explain the Ontology's core building blocks: object types, properties, link types, and action types
- Navigate the Ontology Manager and locate key resources
- Describe how Workshop, OSDK applications, and Functions consume the Ontology
- Create your first Ontology object types in your own Developer Tier account
- Understand the coding standards and sprint structure for the rest of the engagement

2. Foundry Platform Architecture

Foundry is organized in layers. Each layer builds on the one below it, and every application you build in this engagement touches several of these layers at once.

2.1 Data Integration

Source systems, files, and databases are connected into Foundry as datasets through Data Connection, and refined through Pipeline Builder or code-based transforms. This is the raw material the rest of the platform operates on.

2.2 The Ontology

The Ontology sits above your datasets and turns raw rows and columns into a semantic model of your business: object types, properties, link types, and action types. Palantir describes the Ontology as an operational layer that acts as a digital twin of an organization, connecting digital assets to their real-world counterparts.

2.3 Functions & Automation

Functions hold your business logic — validation rules, calculations, event handlers, and scheduled jobs. Functions read from and write to the Ontology, and are the mechanism by which REST APIs, notifications, and scheduled polling are implemented later in this engagement.

2.4 Applications

Workshop and OSDK-powered applications (React, Python, Java) are how people and other systems interact with the Ontology day to day. Workshop is a low-code builder; OSDK applications are custom-coded and give full control over the experience.

3. The Ontology: Core Concepts

An Ontology is a categorization of the world. In Foundry, the Ontology maps datasets and models onto four core concepts:

- **Object Type:** defines an entity or event in your business — for example, Vehicle or Transponder
- **Property:** defines one characteristic of an object type — for example, status, make, or issueDate
- **Link Type:** defines a relationship between two object types — for example, a Vehicle is linked to the Transponder installed on it
- **Action Type:** defines how an object can be created, updated, or deleted — every write to the Ontology goes through an action, never a direct write

3.1 A Familiar Analogy

If you are coming from a data engineering background, this mapping may help:

Dataset World	Ontology World
Dataset	Object Type
Row	Object (an instance of the object type)
Column	Property
Foreign key relationship	Link Type

Preview: Every write to the Ontology happens through an Action Type. We will define our first custom action types in Week 2 (Advanced Ontology Modeling) — today we focus on modeling the objects and their relationships.

4. The Ontology Manager Interface

Ontology Manager is the primary UI for building and browsing the Ontology. You can open it directly by adding `/workspace/ontology` to the end of your Foundry home page URL.

4.1 Layout

- **Top bar:** search for Ontology resources, create new resources, and navigate or create branches
- **Sidebar:** navigate to different resources, pages, and applications within Ontology Manager
- **Discover view:** a customizable landing page; if you are new to an Ontology, it highlights recently modified and prominent object types

4.2 Object Type View

Selecting an object type opens its detail view, including an Overview page listing its properties. Selecting a property from that list opens the property editor, where you configure its data type, display name, and constraints.

5. Workshop — Operational Applications on the Ontology

Workshop is Foundry's low-code application builder. Widgets — tables, forms, maps, charts, buttons — bind directly to Ontology object types and action types, so a working operational tool can be assembled without writing application code.

In this engagement, Workshop is used for the Inventory, Shipment handling, Device location, and Check-in/Check-out screens introduced in Phase 2. Today's hands-on exercise includes an optional first look at building a simple Workshop view.

6. Developer Tooling: OSDK, Developer Console & Foundry CLI

6.1 Ontology SDK (OSDK)

The OSDK is a generated, typed client library (TypeScript, Python, or Java) that gives application code ergonomic, high-scale access to the Ontology — reading and writing objects with minimal boilerplate, while Foundry's governance and permissioning are enforced underneath.

6.2 Developer Console

Developer Console is where OSDK applications and OAuth clients are created and managed. It generates API documentation (TypeScript, Python, and cURL) specific to the object types, action types, and functions your application uses.

6.3 Foundry CLI & Code Repositories

Code Repositories host the source for Functions and applications, with version control and review built in. The Foundry CLI lets you develop locally and publish back to your enrollment.

6.4 SuperRepo

SuperRepo is Foundry's newest application pattern: it lets you define Ontology object types, links, and actions as code in the same repository as your Functions and frontend, with the OSDK regenerated automatically whenever those definitions change. Existing Ontology Manager resources can be imported into a SuperRepo instead of being redefined, so applications are never siloed by how their Ontology types were originally created. This engagement primarily uses the Ontology Manager UI for modeling, with SuperRepo introduced as an advanced option in later weeks.

7. React + OSDK Architecture Preview

Starting Day 2, you will build a React Progressive Web Application that talks to the Ontology through a generated OSDK client:

React PWA → OSDK Client (generated) → Ontology (Objects & Actions)

The OSDK client is generated specifically for your Ontology's object types and actions, so the React application interacts with strongly typed objects such as Vehicle and Transponder rather than raw API calls.

8. Coding Standards for This Engagement

- Property naming: lowerCamelCase (e.g. vehicleId, currentLocation)
- Object type display names: PascalCase, singular (e.g. Vehicle, Transponder)
- TypeScript for all React/OSDK application code; Python for Functions and data transforms
- One Code Repository per application; Ontology changes are reviewed before publishing
- Every Pull Request references its Sprint and includes a short description of the Ontology or UI change
- Prefer small, reviewable commits over large batch commits

9. Sprint Plan Recap

Everything built in Phase 1 becomes the foundation for the five guided sprints in Phase 2:

Sprint	Focus
Sprint 1	Ontology — Shipment, Inventory, Transponder, Bin
Sprint 2	React App — Add / Remove Transponder
Sprint 3	Shipment APIs — external integrations, polling
Sprint 4	Barcode, Inventory & Geospatial Visualization
Sprint 5	Testing, Deployment, Security

10. Key Terms Glossary

Term	Definition
Ontology	The semantic layer mapping datasets and models to object types, properties, link types, and action types.
Object Type	Defines an entity or event, e.g. Vehicle.
Property	A characteristic of an object type, e.g. status.
Link Type	A relationship between two object types.
Action Type	Defines how an object can be created, updated, or deleted.
Ontology Manager	The UI for building and browsing the Ontology, at /workspace/ontology.
OSDK	Ontology Software Development Kit — a generated, typed client library for reading and writing Ontology objects from code.
Developer Console	The application used to create and manage OSDK applications and OAuth clients.
Foundry CLI	Command-line tool for local development and publishing to your enrollment.
Code Repository	Version-controlled home for Functions and application source code.
SuperRepo	A repository pattern that defines Ontology, Functions, and frontend together, generating the OSDK from code.
Workshop	Foundry's low-code application builder for operational tools.
Object Storage V2	A modern backing store for Ontology objects, without the 10,000-result paging limit older backends impose.

11. Reference Documentation

These official Palantir Foundry documentation pages were used as the basis for today's material and are recommended further reading:

- Ontology — Core concepts: palantir.com/docs/foundry/ontology/core-concepts
- Ontology — Overview: palantir.com/docs/foundry/ontology/overview
- Object backend — Overview: palantir.com/docs/foundry/object-backend/overview
- Ontology Manager — Overview: palantir.com/docs/foundry/ontology-manager/overview
- Ontology Manager — Navigation: palantir.com/docs/foundry/ontology-manager/navigation
- Ontology SDK — Overview: palantir.com/docs/foundry/ontology-sdk/overview
- Getting started — Overview: palantir.com/docs/foundry/getting-started/overview
- Foundry announcements (SuperRepo): palantir.com/docs/foundry/announcements/2026-08

Keeping current: Foundry's UI and documentation evolve continuously. Always check the in-platform documentation for the most current screens and terminology before each session.

12. What's Next — Day 2 Preview

Module 1: OSDK Architecture

- OSDK overview and SDK concepts
- Architecture and Object APIs
- Actions and Queries
- Authentication

Hands-on lab: develop your first OSDK application.